Ticket 5

TCK05-MapDrive

Summary

Category	Group Policy / file access
Request	Finance staff need the Finance share to appear automatically as a drive letter
Action	Build a Group Policy Preference that maps `F:` to the Finance share, targeted to `GRP-Finance`
Tools	Group Policy Management (GPMC), Group Policy Management Editor, Command Prompt, File Explorer
Outcome	Resolved. A Finance user gets `F:` at sign-in and a Sales user doesn't

Problem

"My Finance drive isn't showing." Finance staff need the Finance share to appear automatically as a drive letter.

Background. A **mapped drive** is a shared folder on a server given a drive letter on the user's PC. The share lives at a network path (`\\WIN-4G8SQK9BMC1\Finance`), and mapping it makes it appear as `F:` in File Explorer. Group Policy can create the mapping automatically at sign-in, so users never set it up by hand. "My drive isn't showing" means that mapping didn't happen.

In many companies this policy already exists and the helpdesk job is **troubleshooting** it. Here the policy is built first, tested, and then the troubleshooting flow is written down (see Troubleshooting Steps).

Term	Meaning
GPO	A set of settings stored in AD that applies to users and computers
Preference	A Group Policy item that configures something (like a drive) without locking it down
Link	Attaching a GPO to a domain or OU, which decides *whose* policy it is
Item-level targeting	A filter on a single preference item, which decides *which of those users* actually gets the drive

The link answers "who processes this GPO?". Targeting answers "who gets this particular drive?".

Environment

Item	Detail
Domain	acme.local
GPO	`Map-Finance-Drive`
Share	`\\WIN-4G8SQK9BMC1\Finance`
Drive letter	`F:`
Targeted group	`GRP-Finance`
Test users	`dhaddad` (Finance), `acarter` (Sales, `GRP-Sales` only)
Tools	GPMC, Group Policy Management Editor, Command Prompt, File Explorer

Troubleshooting Steps

Build and test

#	Action	Result
1	In GPMC, right-click **Group Policy Objects** → **New** and name it `Map-Finance-Drive`. A new GPO does nothing until it is configured and linked.	GPO created.
2	Right-click the GPO → **Edit**, then **User Configuration** → **Preferences** → **Windows Settings** → **Drive Maps**, right-click → **New** → **Mapped Drive**. Drive maps live under **User** Configuration because they belong to the person, not the PC.	New Drive Properties dialog opens.
3	Set Action **Create**, Location `\\WIN-4G8SQK9BMC1\Finance`, **Reconnect** ticked, Drive letter `F:`.	Drive defined (see settings table below).
4	**Common** tab: tick **Item-level targeting** → **Targeting...** → New Item → **Security Group** → `GRP-Finance`, set to **User in group**.	Only users in that group receive the drive.
5	Link the GPO at the domain root (`acme.local`), link enabled. Security filtering left at the default, **Authenticated Users**.	GPO linked.
6	Sign out the previous session first (drive maps are user settings applied at sign-in). Sign in as Finance user dhaddad. Run `whoami`, `whoami /groups`, `gpupdate /force`.	`ACME\GRP-Finance` is listed; both Computer and User policy update successfully.
7	Check *This PC*.	Finance share appears under *Network locations* as `F:`.
8	Negative test: sign in as `acarter` (`GRP-Sales` only).	No `F:` drive, even though the same GPO applies domain-wide.

Setting	Value	Why
Action	**Create**	Create the mapping if it doesn't exist
Location	`\\WIN-4G8SQK9BMC1\Finance`	The share's network path
Reconnect	Ticked	Restore the drive at each sign-in
Drive letter	`F:`	The letter users will see

Scope note: a domain-level link means every user processes this GPO. That is harmless here, because item-level targeting limits the mapping itself to `GRP-Finance`. The usual practice is to link as close to the affected users as possible, such as the OU that holds them.

Troubleshooting: "my drive isn't showing". Work through these in order and stop at the first failure:

#	Check	How	If it fails
1	Is the user in the right group?	`whoami /groups` shows `GRP-Finance`	Add the user (Ticket 04), then sign out and back in
2	Has policy refreshed?	`gpupdate /force`, then sign out and in	Drive maps apply at sign-in
3	Is the GPO reaching the user?	`gpresult /r /scope user` lists `Map-Finance-Drive` under applied GPOs	Check the link and that the user's account is in the linked scope
4	Is the GPO linked and enabled?	GPMC scope tab, or `Get-GPInheritance`	Link it, or enable the link
5	Can the PC reach the server?	`Test-NetConnection WIN-4G8SQK9BMC1 -Port 445`, or open `\\WIN-4G8SQK9BMC1\Finance` from Run	A DNS or network problem, not a policy problem

#	Check	How	If it fails
6	Does the share allow the group?	`Get-SmbShareAccess -Name Finance`	See Ticket 04
7	Is the drive letter free?	Look for another device already using `F:`	Choose another letter in the policy

Reading gpresult: a user who is filtered out by item-level targeting still sees the GPO listed as applied, because the filtering happens inside the preference item. So when the GPO is listed but the drive is missing, check **group membership first**.

Root Cause

The mapping didn't exist: no policy created the drive for Finance users. The policy is built so that the link decides who processes the GPO, and item-level targeting decides which of those users actually gets the drive. For a Sales user outside `GRP-Finance`, no drive is the correct result.

Fix

Create the `Map-Finance-Drive` GPO with a mapped drive preference (`F:` to `\\WIN-4G8SQK9BMC1\Finance`, Create, Reconnect), target it to `GRP-Finance` with item-level targeting, and link it at `acme.local` (Troubleshooting Steps, steps 1 to 5).

Verification

Finance user (`dhaddad`): `ACME\GRP-Finance` is listed in `whoami /groups`, both Computer and User policy update successfully, and the Finance share appears under *Network locations* as `F:`.

User outside Finance (`acarter`): in `GRP-Sales` only. There is no `F:` drive under *This PC*, even though the same GPO applies domain-wide. That is item-level targeting doing its job.

Lessons Learned

• **Group membership changes need a new sign-in.** A user added to `GRP-Finance` won't get the drive until they sign out and back in, because their token is built at sign-in.

• **Ticket 04 interaction:** Bob Nasser was added to `GRP-Finance` in that ticket, so he now receives `F:` as well. That is why a Sales user outside the group (`acarter`) was used for the negative test.

• **Building GPOs is usually a sysadmin task.** At helpdesk level the common work is steps 1 to 7 of the troubleshooting table.

Command reference

```
# On the client, in the user's session
whoami /groups
gpupdate /force
gpresult /r /scope user

# On the server
Get-GPO -Name Map-Finance-Drive
Get-GPInheritance -Target "DC=acme,DC=local"
Get-SmbShareAccess -Name Finance
```

Screenshots

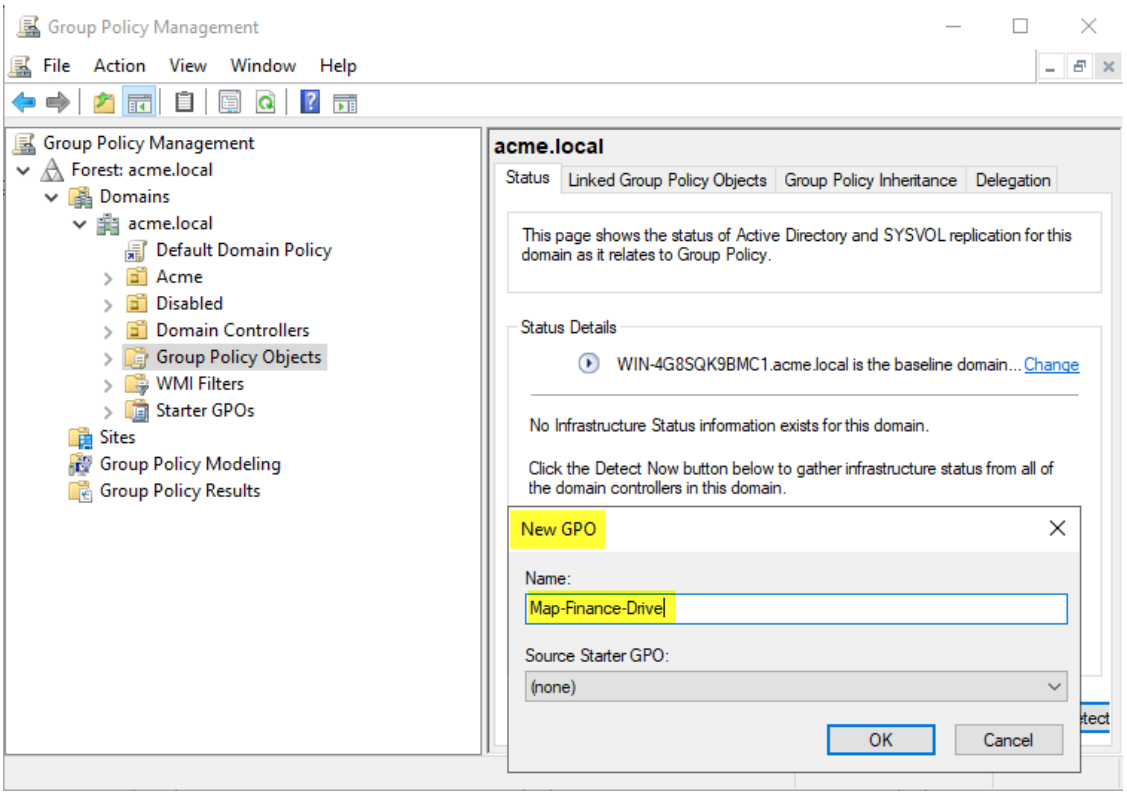


Figure 1. New GPO dialog with the name Map-Finance-Drive

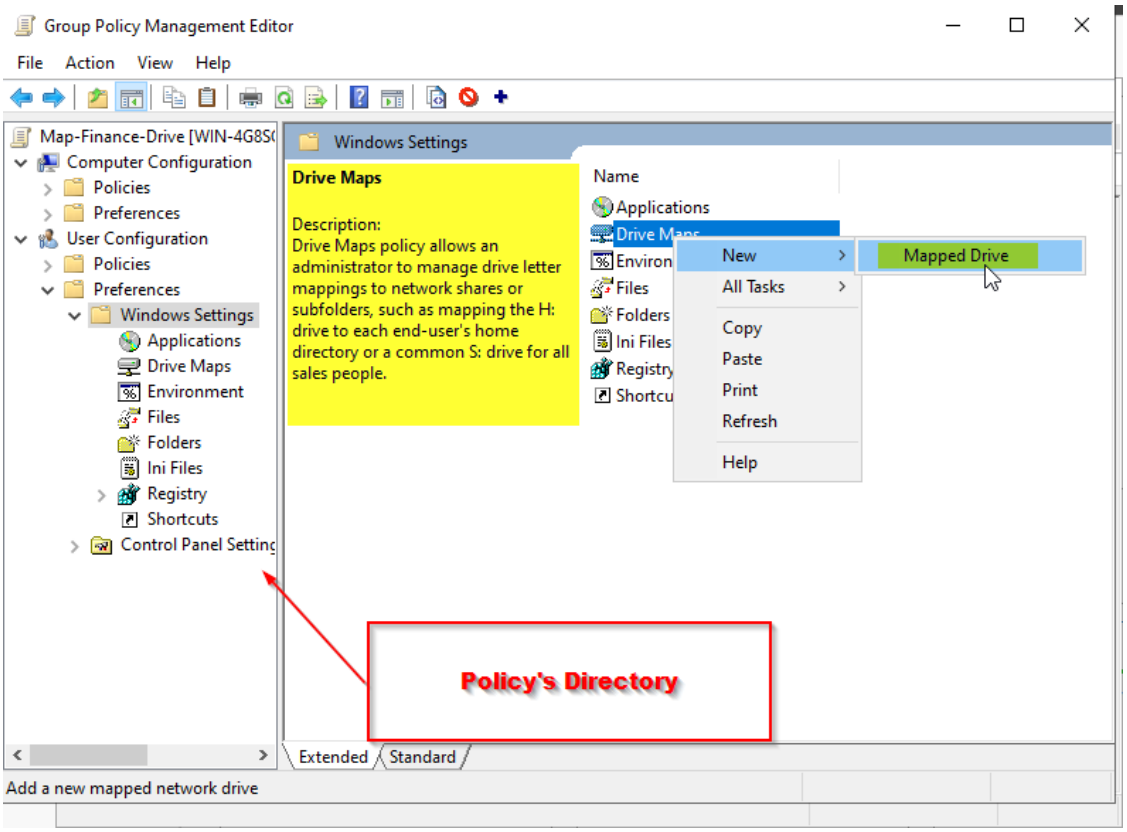


Figure 2. Group Policy editor with Drive Maps selected and New → Mapped Drive

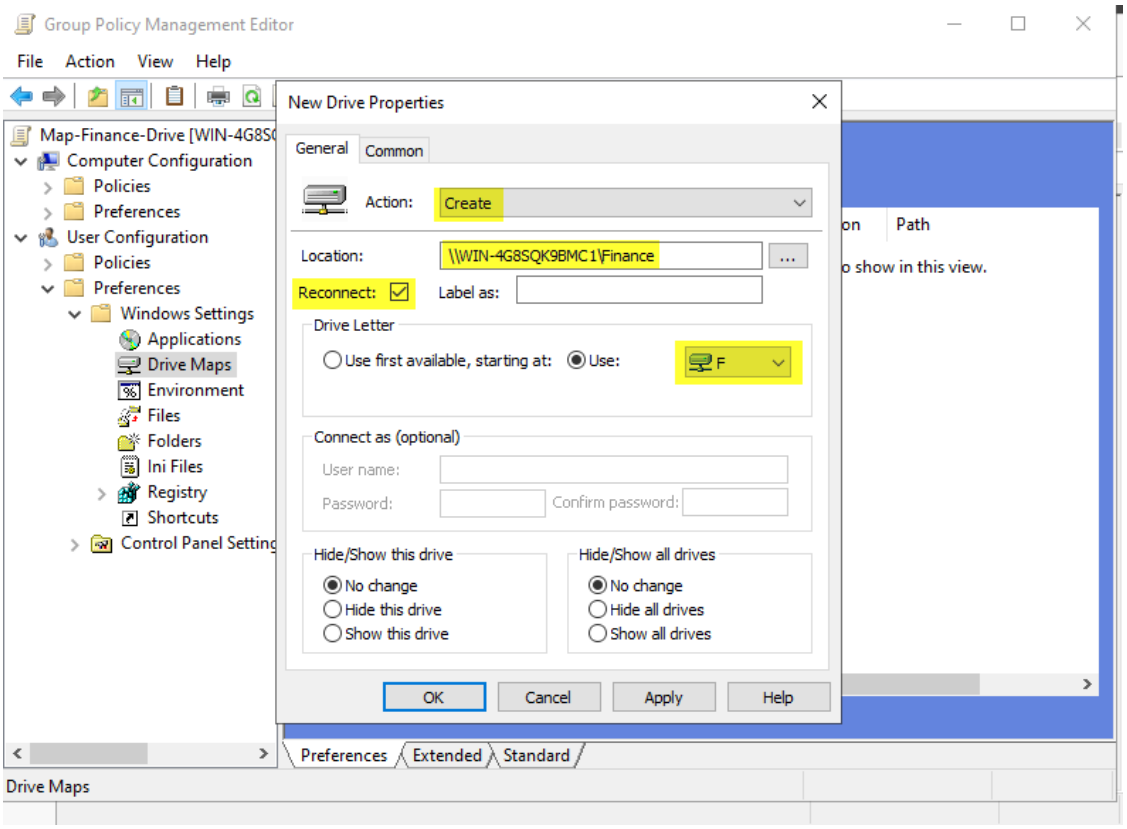


Figure 3. New Drive Properties with Create, the Finance path, Reconnect and F:

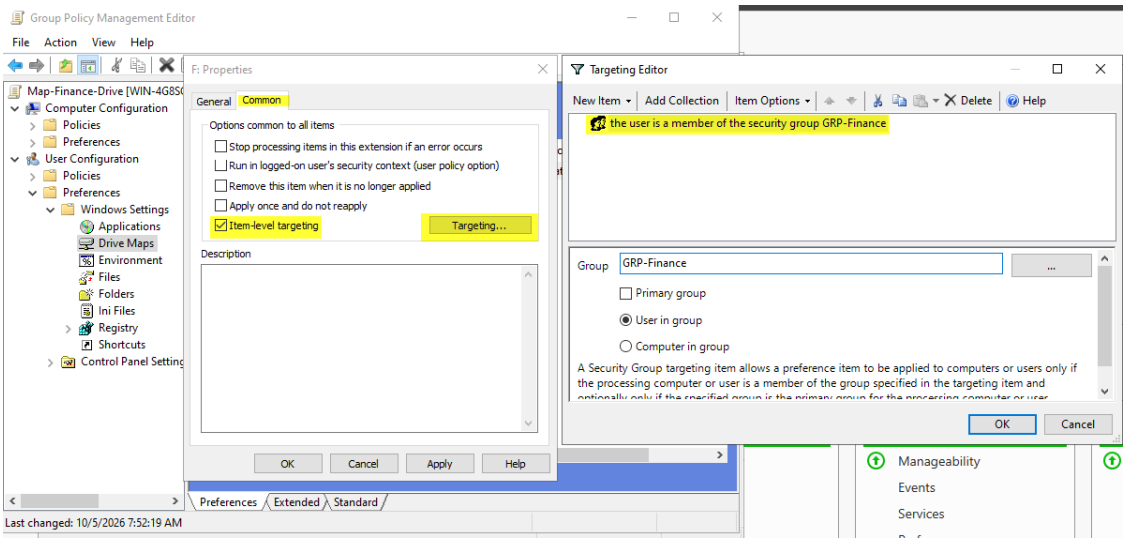


Figure 4. Targeting editor: the user is a member of the security group GRP-Finance

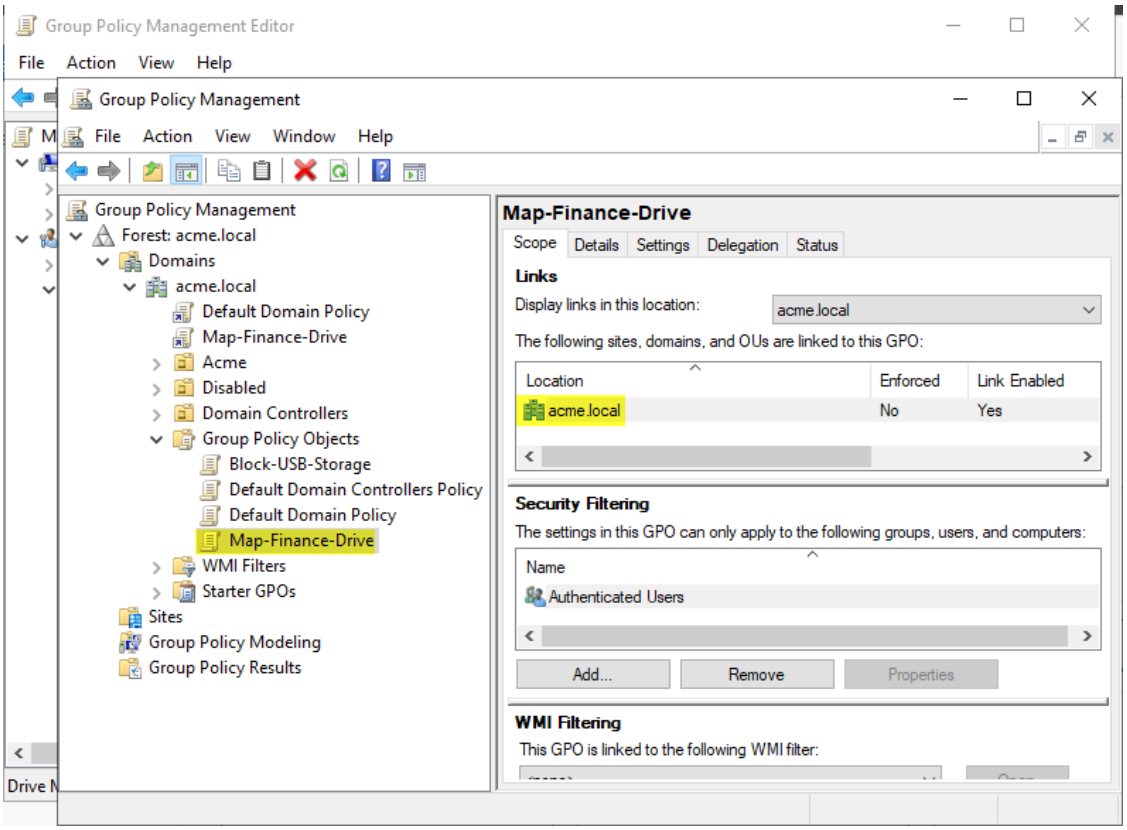


Figure 5. GPMC scope tab showing the GPO linked to acme.local

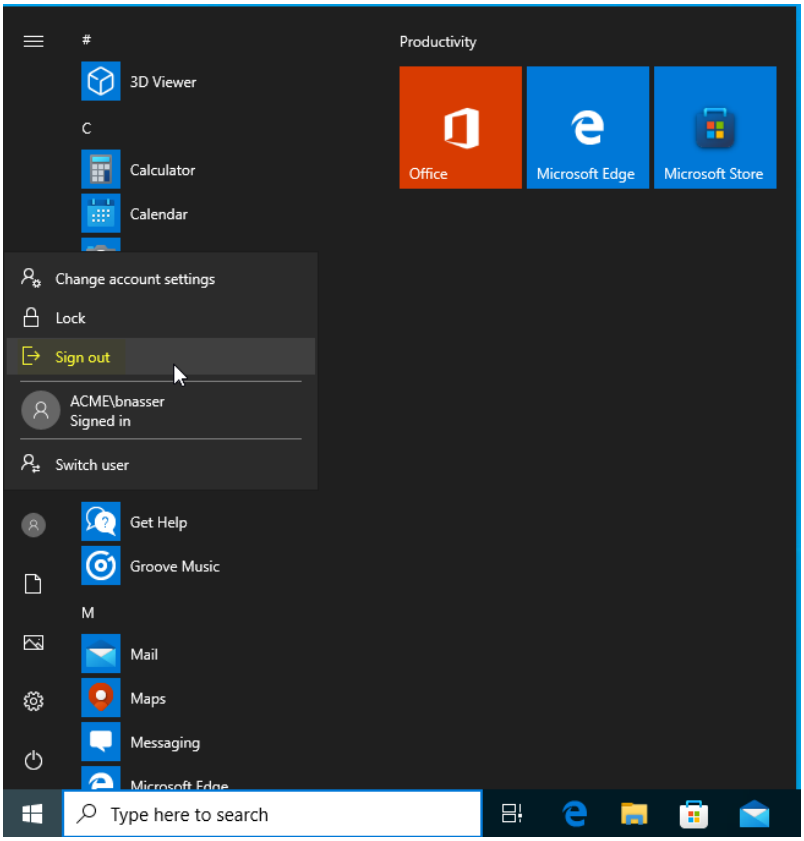


Figure 6. Start menu with Sign out highlighted

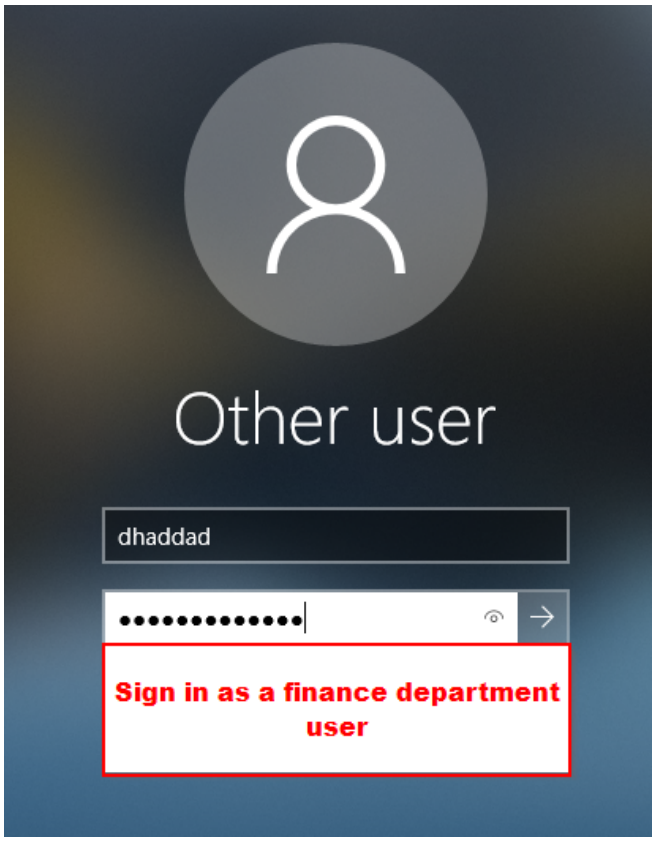


Figure 7. Sign-in screen for dhaddad

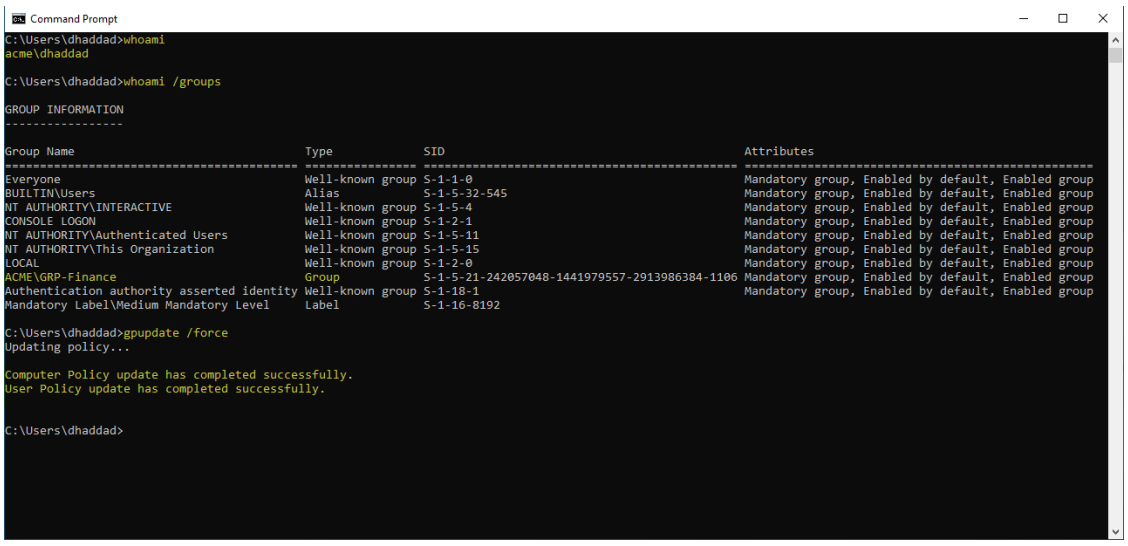


Figure 8. whoami, whoami /groups and gpupdate /force output for dhaddad

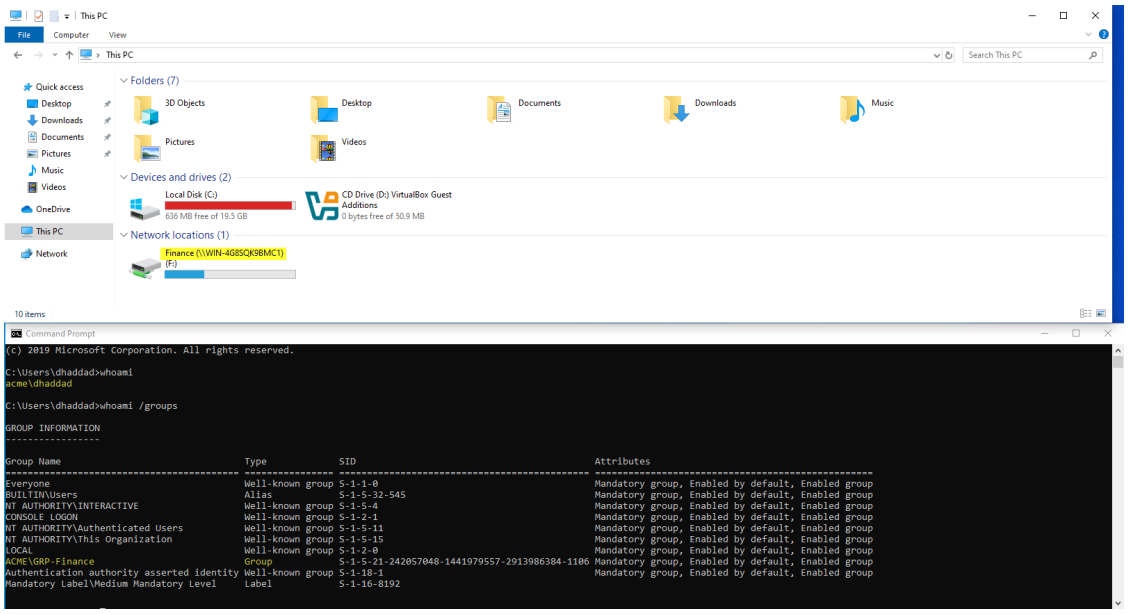


Figure 9. This PC showing Finance (\\WIN-4G8SQK9BMC1) mapped as F:

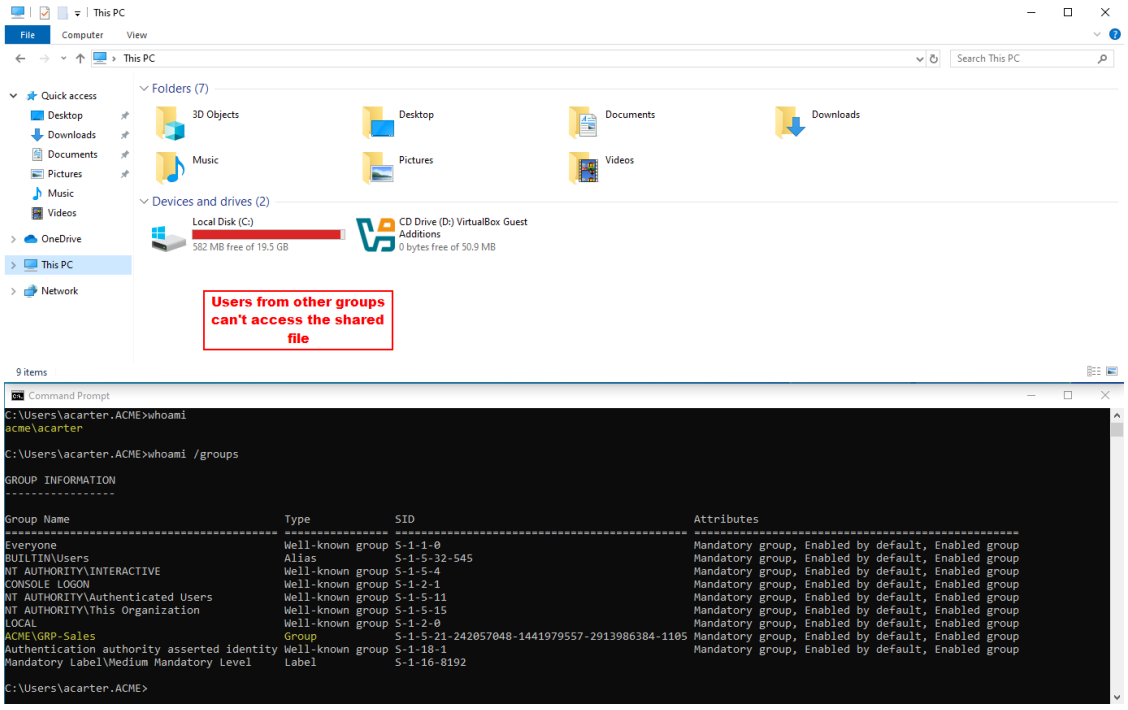


Figure 10. This PC for acarter with no mapped drive, and whoami /groups showing only GRP-Sales